

Symmetry and Equivariance in Reinforcement Learning: Theory for Finite Symmetric MDPs with a Snake Case Study

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Abstract

Many reinforcement-learning environments are symmetric: rotating or reflecting the world leaves the dynamics and rewards unchanged. We study what this structure guarantees and whether standard algorithms exploit it. Working at the level of a finite G -symmetric MDP for an arbitrary finite group G , we show that the optimal value function is G -invariant, that the set of optimal policies is closed under the group action, and consequently that an equivariant optimal policy always exists, so restricting to equivariant policies sacrifices no optimality. We instantiate this framework on the game of Snake under the dihedral group D_4 , verifying both in proof and in code that Snake is exactly D_4 -symmetric. We then introduce an equivariance-error metric and use it to measure how equivariant a learned policy actually is. A standard (non-equivariant) DQN agent does *not* recover the equivariant optimum its environment guarantees: its equivariance error grows from a near-uniform initialization and plateaus well above zero, spread roughly evenly across the group and concentrated in harder, longer-snake states. A D_4 -equivariant network, trained identically, instead remains exactly equivariant throughout training while learning faster and reaching higher performance with roughly $21\times$ fewer parameters. The existence of an equivariant optimum and the behavior of a symmetry-blind learner are thus distinct, and equivariant architectures are what bridges them, most cleanly when the symmetry is exact, as in Snake.

1 Introduction

Many of the environments studied in reinforcement learning (RL) are symmetric. Board games, gridworlds, and a great many physical control tasks are invariant under rotations or reflections of space: if the entire configuration is rotated by 90, the dynamics and rewards transform with it, and the appropriate behavior is simply the rotated version of the original behavior. An agent that has learned to act well in one orientation should not have to relearn the task from scratch in another. This intuition, that the solution should respect the symmetry of the problem, is the same one that motivates equivariant deep learning, where building a known symmetry into the network as an inductive bias improves data efficiency and generalization [3, 4]. It also connects to a line of work on symmetric MDPs and MDP homomorphisms that formalizes symmetry as state-action abstraction [1, 2].

At the level of optimal solutions, symmetry has a clean and strong consequence. For *any* finite group G acting on a finite MDP whose transition kernel and reward are invariant under the group,

an equivariant optimal policy exists: one can always choose an optimal policy that responds to a transformed state with the correspondingly transformed action distribution. Restricting the search to equivariant policies therefore costs nothing in optimality. We give self-contained statements and proofs of this and the supporting facts, namely invariance of the optimal value function and the orbit structure of optimal policies, at full generality for an arbitrary finite group rather than a specific one.

This existence guarantee, however, is silent about *learning*. It says an equivariant optimal policy is available; it does not say that an RL algorithm training an ordinary, symmetry-blind neural network will find one, or even approach one. Standard architectures are free to represent equivariant policies but are in no way required to, and the training objective rewards only return, not symmetry. Whether a learned policy actually becomes equivariant is thus an empirical question, and the gap between what is guaranteed to exist and what is actually recovered is the central subject of this paper.

To study that gap concretely we use the game of Snake under the dihedral group D_4 , the eight-element symmetry group of the square. Snake is a convenient testbed precisely because its symmetry is *exact*: on a square grid with uniform food respawning, the dynamics and rewards are exactly D_4 -invariant (we prove this and verify it numerically), with none of the boundary asymmetries or direction-dependent rewards that complicate other environments. It is also discrete, easy to visualize, and nontrivial to play well, which lets us inspect not just aggregate symmetry statistics but the specific states on which a policy’s symmetry breaks down.

We organize the empirical study around three questions. (RQ1) Does a standard DQN agent become more equivariant as it learns to play, and does it ever reach the equivariance its environment guarantees to be attainable? (RQ2) If not, how does its asymmetry fail: uniformly across the symmetry group, or concentrated on particular elements and states? (RQ3) Does an architecturally equivariant agent close the gap, and what does it cost? Our answers are, respectively: no, the equivariance error *grows* from initialization and plateaus well above zero; the failure is roughly isotropic across D_4 and concentrated in harder, longer-snake configurations; and a D_4 -equivariant network is exactly equivariant throughout training while learning faster, playing better, and using far fewer parameters.

Contributions. This paper makes four contributions. First, we develop the theory of finite G -symmetric MDPs for an arbitrary finite group and prove, in self-contained form, the invariance of the optimal value (Lemma 1), the orbit structure of optimal policies (Theorem 2), and the existence of an equivariant optimal policy (Corollary 3). Second, we instantiate the framework on Snake, giving an explicit D_4 action on its states and actions and proving (and numerically verifying) that the resulting MDP is exactly D_4 -symmetric (Proposition 4). Third, we introduce an equivariance-error metric for policies and use it to show that a standard DQN agent does not recover the equivariant optimum, characterizing how and where its symmetry breaks (Section 6). Fourth, we show that a D_4 -equivariant network trained under an identical configuration is exactly equivariant by construction and is simultaneously more sample-efficient and more performant at a fraction of the parameter count (Section 7).

Road map. Section 2 fixes notation for MDPs, group actions, and D_4 . Section 3 develops the general theory of symmetric MDPs and equivariant policies. Section 4 instantiates it on Snake and establishes D_4 -symmetry. Section 5 describes the agents and training setup, Section 6 measures equalvariance in the learned baseline, and Section 7 compares against an equivariant architecture. ?? discusses generalization beyond Snake, and Section 8 discusses implications, limitations, and

the exact-versus-approximate-symmetry trade-off that determines when equivariant architectures are worth their cost.

2 Preliminaries

This section establishes notation and reviews the mathematical objects used throughout the paper: Markov decision processes, group actions, and the dihedral group D_4 , which serves as our running concrete example.

2.1 Markov Decision Processes

A finite, discounted Markov decision process (MDP) is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ where $\mathcal{S}$ is a finite set of states, $\mathcal{A}$ a finite set of actions, $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ a transition kernel with $\sum_{s'} P(s' | s, a) = 1$, $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ a reward function, and $\gamma \in [0, 1)$ a discount factor.

A (stationary, stochastic) policy is a map $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$, where $\Delta(\mathcal{A})$ denotes the set of probability distributions over $\mathcal{A}$. The value of a policy from state s is

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s \right]. \quad (1)$$

A policy π^* is optimal if $V^{\pi^*}(s) \geq V^\pi(s)$ for all s and all π . For finite MDPs, an optimal policy exists; in general it need not be unique.

2.2 Group Actions

Let G be a finite group with identity e and X a set. A (left) group action of G on X is a map $G \times X \rightarrow X$, written $(g, x) \mapsto g \cdot x$, satisfying:

1. $e \cdot x = x$ for all $x \in X$,
2. $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$.

If G acts on both X and Y , a map $f : X \rightarrow Y$ is *equivariant* if $f(g \cdot x) = g \cdot f(x)$ for all $g \in G, x \in X$. It is *invariant* if $f(g \cdot x) = f(x)$ for all g, x (equivalently, equivariant when the action on Y is trivial).

2.3 The Dihedral Group D_4

The dihedral group D_4 is the symmetry group of the square. It has eight elements: four rotations $\{e, r, r^2, r^3\}$ where r denotes rotation by 90 counterclockwise, and four reflections $\{s, sr, sr^2, sr^3\}$ where s is reflection across a fixed axis. The defining relations are $r^4 = e$, $s^2 = e$, and $sr s = r^{-1}$.

D_4 acts naturally on $\mathbb{Z}^2$ by linear transformations: r rotates a coordinate (x, y) by 90, and s reflects across (say) the x -axis. This action restricts to any D_4 -invariant subset of $\mathbb{Z}^2$, in particular to a square grid centered appropriately. We use D_4 as the concrete group in the Snake case study of Section 4; the theory of Section 3, however, is developed for an arbitrary finite group.

3 Symmetric MDPs and Equivariant Policies

This section develops the theory at full generality: a finite MDP equipped with the action of an arbitrary finite group G whose transition kernel and reward are invariant. Nothing in the definitions

or proofs is specific to Snake or to D_4 ; the Snake formalization of Section 4 is one instance. All three results hold for every finite G -symmetric MDP: invariance of the optimal value (Lemma 1), the orbit structure of optimal policies (Theorem 2), and the existence of an equivariant optimal policy (Corollary 3).

3.1 Group Actions on States, Actions, and Policies

Let G be a finite group acting on the state space $\mathcal{S}$ and on the action space $\mathcal{A}$ of a finite MDP $\mathcal{M}$. (Either action may be trivial; the interesting cases have both nontrivial.) We extend these to an action on policies as follows.

First, G acts on distributions over actions: for $\mu \in \Delta(\mathcal{A})$ and $g \in G$, define $g \cdot \mu \in \Delta(\mathcal{A})$ by

$$(g \cdot \mu)(a) = \mu(g^{-1} \cdot a). \quad (2)$$

This is the pushforward of μ under the permutation $a \mapsto g \cdot a$, and it is a valid left action on $\Delta(\mathcal{A})$. The induced action on policies $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ is

$$(g \cdot \pi)(s)(a) := \pi(g^{-1} \cdot s)(g^{-1} \cdot a). \quad (3)$$

That is, the transformed policy at the original state takes the original action with probability equal to the original policy at the transformed state taking the transformed action. A policy is *equivariant* if $g \cdot \pi = \pi$ for all $g \in G$, equivalently

$$\pi(g \cdot s)(g \cdot a) = \pi(s)(a) \quad \forall g \in G, s \in \mathcal{S}, a \in \mathcal{A}. \quad (4)$$

Intuitively, an equivariant policy responds to a transformed state with the correspondingly transformed action distribution: if the world is rotated, the recommended move rotates with it.

3.2 Symmetric MDPs

Definition 1 (G -symmetric MDP). Let G be a finite group acting on $\mathcal{S}$ and on $\mathcal{A}$. The MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ is G -symmetric if for all $g \in G$, all $s, s' \in \mathcal{S}$, and all $a \in \mathcal{A}$,

$$P(g \cdot s' \mid g \cdot s, g \cdot a) = P(s' \mid s, a), \quad (5)$$

$$R(g \cdot s, g \cdot a) = R(s, a). \quad (6)$$

Equivalently, transforming the current state, action, and successor state simultaneously by the same group element leaves the transition probabilities and reward unchanged: the dynamics and reward are blind to the group's action. This is the setting of symmetric MDPs and MDP homomorphisms studied by [1, 2]; we record the consequences we need in self-contained form.

3.3 Invariance of the Optimal Value

Lemma 1 (Invariance of optimal value). *Let $\mathcal{M}$ be a finite G -symmetric MDP. Its optimal state-value function V^* and optimal action-value function Q^* satisfy, for all $g \in G$,*

$$V^*(g \cdot s) = V^*(s) \quad \forall s \in \mathcal{S}, \quad (7)$$

$$Q^*(g \cdot s, g \cdot a) = Q^*(s, a) \quad \forall s \in \mathcal{S}, a \in \mathcal{A}. \quad (8)$$

Proof. The optimal value functions are the unique fixed points of the Bellman optimality operators $\mathcal{T}^*$ on the spaces of bounded functions on $\mathcal{S}$ (resp. $\mathcal{S} \times \mathcal{A}$); uniqueness follows from $\gamma < 1$, which makes $\mathcal{T}^*$ a contraction. It therefore suffices to show that $\mathcal{T}^*$ maps the subspace of G -invariant functions into itself: its unique fixed point must then be invariant.

For Q^* , the Bellman operator is

$$(\mathcal{T}^*Q)(s, a) = R(s, a) + \gamma \sum_{s'} P(s' | s, a) \max_{a'} Q(s', a'). \quad (9)$$

Let Q be G -invariant, i.e. $Q(g \cdot s, g \cdot a) = Q(s, a)$ for all g, s, a . Evaluating $\mathcal{T}^*Q$ at $(g \cdot s, g \cdot a)$ and using Definition 1,

$$(\mathcal{T}^*Q)(g \cdot s, g \cdot a) = R(g \cdot s, g \cdot a) + \gamma \sum_{s'} P(s' | g \cdot s, g \cdot a) \max_{a'} Q(s', a') \quad (10)$$

$$= R(s, a) + \gamma \sum_{s''} P(g \cdot s'' | g \cdot s, g \cdot a) \max_{a'} Q(g \cdot s'', a') \quad (\text{reindex } s' = g \cdot s'') \quad (11)$$

$$= R(s, a) + \gamma \sum_{s''} P(s'' | s, a) \max_{a'} Q(g \cdot s'', a'). \quad (12)$$

In the inner maximization, as a' ranges over $\mathcal{A}$ so does $a' = g \cdot a''$ (the action of g is a bijection on $\mathcal{A}$), and by invariance of Q we have $Q(g \cdot s'', a') = Q(g \cdot s'', g \cdot a'') = Q(s'', a'')$; hence $\max_{a'} Q(g \cdot s'', a') = \max_{a''} Q(s'', a'')$. Therefore

$$(\mathcal{T}^*Q)(g \cdot s, g \cdot a) = R(s, a) + \gamma \sum_{s''} P(s'' | s, a) \max_{a''} Q(s'', a'') = (\mathcal{T}^*Q)(s, a), \quad (13)$$

so $\mathcal{T}^*Q$ is G -invariant. The unique fixed point Q^* is thus invariant, and the result for V^* follows from $V^*(s) = \max_a Q^*(s, a)$ together with invariance of Q^* . $\square$

3.4 Orbit Structure and Existence of an Equivariant Optimum

Theorem 2 (Orbit structure of optimal policies). *Let $\mathcal{M}$ be a finite G -symmetric MDP. If π^* is an optimal policy, then $g \cdot \pi^*$ is also an optimal policy for every $g \in G$.*

Proof. Optimality of a policy π is equivalent to the support condition: for every state s , $\pi(s)$ assigns positive probability only to actions a achieving $Q^*(s, a) = V^*(s)$.

Suppose π^* is optimal. Fix $g \in G$ and a state s , and let a be any action with $(g \cdot \pi^*)(s)(a) > 0$. By definition $(g \cdot \pi^*)(s)(a) = \pi^*(g^{-1} \cdot s)(g^{-1} \cdot a)$, so $\pi^*(g^{-1} \cdot s)(g^{-1} \cdot a) > 0$, which by optimality of π^* forces $Q^*(g^{-1} \cdot s, g^{-1} \cdot a) = V^*(g^{-1} \cdot s)$. By Lemma 1, $Q^*(g^{-1} \cdot s, g^{-1} \cdot a) = Q^*(s, a)$ and $V^*(g^{-1} \cdot s) = V^*(s)$, so $Q^*(s, a) = V^*(s)$. Thus $(g \cdot \pi^*)(s)$ is supported on $\arg \max_a Q^*(s, a)$ for every s , i.e. $g \cdot \pi^*$ is optimal. $\square$

Corollary 3 (Existence of an equivariant optimal policy). *Every finite G -symmetric MDP admits an optimal policy π^* that is G -equivariant in the sense of Equation (4).*

Proof. Let π_0^* be any optimal policy (one exists since $\mathcal{S}, \mathcal{A}$ are finite). Define the group average

$$\pi^*(s) := \frac{1}{|G|} \sum_{g \in G} (g \cdot \pi_0^*)(s). \quad (14)$$

Each $g \cdot \pi_0^*$ is optimal by Theorem 2, and the set of optimal policies is convex: any mixture of policies whose per-state supports lie in $\arg \max_a Q^*(s, \cdot)$ again has support in that set. Hence π^* is optimal.

For equivariance, fix $h \in G$. Using the policy action and associativity,

$$(h \cdot \pi^*)(s) = \frac{1}{|G|} \sum_{g \in G} (h \cdot (g \cdot \pi_0^*))(s) = \frac{1}{|G|} \sum_{g \in G} ((hg) \cdot \pi_0^*)(s). \quad (15)$$

Since $g \mapsto hg$ is a bijection of G , the sum is unchanged, so $(h \cdot \pi^*)(s) = \pi^*(s)$ for all s . Thus π^* is equivariant. $\square$

Corollary 3 is the theoretical motivation for the rest of the paper: since an equivariant optimal policy always exists in a symmetric MDP, one can restrict the search to equivariant policies without sacrificing optimality. This is exactly what equivariant network architectures do by construction.

Remark 1 (Scope of the assumptions). Two assumptions are essential. (i) *Exact symmetry*: Definition 1 requires P and R to be *exactly* invariant; environments that are only approximately symmetric (e.g. broken by boundary effects, asymmetric goals, or gravity) admit only approximately equivariant optima, with the gap controlled by the size of the symmetry violation. (ii) *Finiteness*: we use a finite group (the averaging in Corollary 3 is a finite sum) and a finite MDP (existence of an optimal policy). Both can be relaxed (compact groups via Haar-measure averaging, and continuous spaces via standard measure-theoretic arguments), but we do not pursue this here. We return to these points in ?? and section 8.

4 Snake as a D_4 -Symmetric MDP

We now instantiate the general framework on a concrete environment: the game of Snake, with the dihedral group $G = D_4$. We formalize Snake as an MDP, define the D_4 action on its states and actions, and verify that it satisfies Definition 1. The general results of Section 3 then apply verbatim, yielding a D_4 -equivariant optimal policy for Snake.

4.1 Game Description

Snake is played on an $N \times N$ grid (we take N even for symmetry, e.g., $N = 8$). At each timestep, a snake of length $L_t \geq 1$ occupies a sequence of grid cells, and a single food item occupies one other cell. The snake has a head (the most recently added cell) and moves in one of four cardinal directions. Each step, the head advances one cell in the chosen direction; if it lands on the food, the snake grows by one and a new food is spawned uniformly at random on an empty cell; otherwise the tail cell is removed. The game terminates when the head leaves the grid or collides with the snake’s own body.

4.2 State Representation

We represent states as multi-channel binary tensors $s \in \{0, 1\}^{C \times N \times N}$ with $C = 3$ channels:

- Channel 1: snake head indicator (1 in the head cell, 0 elsewhere)
- Channel 2: snake body indicator (1 in body cells excluding head, 0 elsewhere)
- Channel 3: food indicator (1 in food cell, 0 elsewhere)

We additionally include a small auxiliary signal indicating the snake’s heading direction; for the purposes of the formal analysis below, this can be appended as a one-hot vector of dimension 4. For brevity we suppress this auxiliary signal in the notation when no confusion arises.

The state space $\mathcal{S}$ is the set of all such tensors arising from valid game configurations.

4.3 Action Space

We use absolute cardinal actions: $\mathcal{A} = \{\text{up, down, left, right}\}$. We discuss the alternative of *relative* actions (turn left, turn right, go straight) in Section 4.7.

4.4 Transition and Reward

The transition $P(\cdot \mid s, a)$ is deterministic in the snake dynamics, with stochasticity coming only from food respawning. The reward function is

$$R(s, a) = \begin{cases} +1 & \text{if action } a \text{ from } s \text{ results in eating food} \\ -1 & \text{if action } a \text{ from } s \text{ results in collision} \\ -c & \text{otherwise} \end{cases} \quad (16)$$

where $c > 0$ is a small step penalty (we use $c = 0.01$).

4.5 The D_4 Action on Snake

We specialize the abstract group action of Section 3.1 to D_4 acting on Snake states and actions.

Action on states. Each element $g \in D_4$ acts on the spatial coordinates of the grid via the corresponding rotation or reflection (Section 2.3). We extend this to states by acting independently on each channel: for $g \in D_4$ and a state tensor $s \in \{0, 1\}^{C \times N \times N}$,

$$(g \cdot s)_{c,i,j} = s_{c,g^{-1}(i,j)}, \quad (17)$$

where $g^{-1}(i, j)$ denotes the spatial coordinates obtained by applying g^{-1} to (i, j) . This “pullback” formulation ensures composition works correctly, $(gh) \cdot s = g \cdot (h \cdot s)$. Geometrically, if g rotates the world by 90, then $g \cdot s$ is the state in which the entire game configuration appears rotated by 90. The action preserves $\mathcal{S}$: the image of a valid Snake state under $g \in D_4$ is again valid, since the rules depend only on the geometric configuration on the grid and D_4 preserves the grid.

Action on actions. The four cardinal directions are vectors in $\mathbb{Z}^2$:

$$\text{up} \leftrightarrow (0, 1), \quad \text{down} \leftrightarrow (0, -1), \quad \text{left} \leftrightarrow (-1, 0), \quad \text{right} \leftrightarrow (1, 0). \quad (18)$$

D_4 acts on these direction vectors by its standard action on $\mathbb{Z}^2$, inducing the action on $\mathcal{A}$:

$$g \cdot a := \text{the unique } a' \in \mathcal{A} \text{ such that the direction vector of } a' = g(\text{direction vector of } a). \quad (19)$$

For example, if r is rotation by 90 counterclockwise then $r \cdot \text{up} = \text{left}$. This is a permutation action of D_4 on the four-element set $\mathcal{A}$. Together with the state action, it induces the action on policies and the equivariance condition Equation (4), now read with $G = D_4$.

4.6 Snake Satisfies the Symmetry Definition

Proposition 4 (Snake is a D_4 -symmetric MDP). *With the actions of Section 4.5, the Snake MDP satisfies Definition 1 for $G = D_4$: for all $g \in D_4$, $s, s' \in \mathcal{S}$, and $a \in \mathcal{A}$, $P(g \cdot s' \mid g \cdot s, g \cdot a) = P(s' \mid s, a)$ and $R(g \cdot s, g \cdot a) = R(s, a)$.*

Proof. The transition function decomposes into the deterministic update of the snake under action a and the stochastic respawning of food when food is eaten; we check both are D_4 -equivariant.

Deterministic snake update. Suppose action a in state s moves the head to an unoccupied cell (no growth, no collision). In state $g \cdot s$ the head sits at the image under g of the original head cell, and action $g \cdot a$ corresponds to the transformed direction vector; by definition of the coordinate action, the resulting head cell is the image under g of the original head cell, and the body update commutes with g . The cases of collision and food consumption are analogous: collision occurs iff the corresponding transformed cell is a wall or body cell, and food is eaten iff the head lands on the transformed food cell.

Food respawning. When food is eaten, a new food cell is sampled uniformly from the unoccupied cells. Since g maps the unoccupied cells of s bijectively to those of $g \cdot s$, the uniform distribution is preserved: the probability of a respawn position f from s equals that of $g \cdot f$ from $g \cdot s$.

Combining the two parts gives $P(g \cdot s' \mid g \cdot s, g \cdot a) = P(s' \mid s, a)$. For rewards: the reward depends only on whether the action results in eating food, collision, or neither; these events are preserved by the group action, so $R(g \cdot s, g \cdot a) = R(s, a)$. $\square$

Corollary 5 (Equivariant optimal policy for Snake). *The Snake MDP admits a D_4 -equivariant optimal policy.*

Proof. Immediate from Proposition 4 and Corollary 3. $\square$

This is the concrete payoff of the general theory: because Snake meets Definition 1, the existence of an equivariant optimum is inherited from Corollary 3 without any further work. The remaining question, whether a standard learning algorithm *recovers* such a policy, is empirical and is the subject of Section 6.

4.7 A Note on the Action Encoding

We chose to encode actions as absolute cardinal directions. An alternative is *relative* actions: $\mathcal{A}' = \{\text{straight, left turn, right turn}\}$, interpreted relative to the snake’s current heading. This changes the formalization.

With relative actions, D_4 does not act on $\mathcal{A}'$ in the same way: rotating the world by 90 does not turn “turn left” into another action, since it remains “turn left” in the snake’s local frame. Instead D_4 acts trivially on $\mathcal{A}'$, while the action on the state must now include the snake’s heading. Under this encoding the symmetry analysis still goes through, but the equivariance condition on policies becomes *invariance*, $\pi(g \cdot s) = \pi(s)$ for all $g \in D_4$ (the special case of Equation (4) with trivial action on $\mathcal{A}$). This is conceptually simpler but requires the state to encode heading explicitly. Both encodings yield valid D_4 -symmetric formulations; we use absolute actions throughout because they integrate more directly with the equivariant-CNN literature, where the input and output transformations are usually both nontrivial.

5 Methods

This section describes the experimental apparatus used to study whether a standard learning algorithm recovers the equivariant optimum guaranteed by Corollary 5. We train value-based agents on the Snake MDP of Section 4 and compare a conventional convolutional network against a D_4 -equivariant one. All design choices below are fixed across the experiments of Sections 6 and 7 unless stated otherwise.

5.1 Environment Implementation

We implement the Snake MDP exactly as formalized in Section 4: an $N \times N$ grid with $N = 8$, the three-channel binary state tensor (head, body, food) of Section 4, the four absolute cardinal actions, and the reward $R(s, a) \in \{+1, -1, -c\}$ with step penalty $c = 0.01$. The four actions are stored as (row, col) direction vectors, and the D_4 action on the grid is applied to those same vectors, so that the action-on-states and action-on-actions are induced by a single linear map; this is what makes the two compatible in the sense required by Proposition 4. Episodes terminate on collision and are additionally truncated at 400 steps to bound trajectory length during training. Reverse moves (a 180 turn into the neck) are illegal and are overridden by the environment to continue straight, as is standard for Snake; the agent’s policy is nonetheless defined over all four absolute actions, which keeps the state a purely spatial D_4 object and the equivariance condition Equation (4) clean.

We verify the symmetry claim of Proposition 4 numerically. Over 300 random transitions and all eight group elements, stepping the transformed state $g \cdot s$ with the transformed action $g \cdot a$ produces exactly the transformed successor $g \cdot s'$ (and we check the group action satisfies $(gh) \cdot s = g \cdot (h \cdot s)$ on both states and actions); all checks pass to machine precision. This confirms the deterministic part of Equation (5) on the implemented environment.

5.2 Learning Algorithm

We use Deep Q-Networks (DQN) [5]. A network $Q_\theta : \mathcal{S} \rightarrow \mathbb{R}^{|A|}$ outputs one value per action; the greedy policy is $\pi_\theta(s) = \arg \max_a Q_\theta(s, a)$, and for the equivariance metric of Section 6 we read off a stochastic policy by taking a softmax over $Q_\theta(s, \cdot)$. Training minimizes the one-step temporal-difference error against a periodically updated target network $Q_{\bar{\theta}}$,

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[\ell_\delta \left(Q_\theta(s, a) - (r + \gamma (1 - \text{done}) \max_{a'} Q_{\bar{\theta}}(s', a')) \right) \right], \quad (20)$$

where $\mathcal{D}$ is an experience-replay buffer and ℓ_δ is the Huber loss. We use discount $\gamma = 0.99$, the Adam optimizer with learning rate 10^{-3} , minibatches of 64 transitions, a replay buffer of capacity 3×10^4 with a warm-up of 10^3 transitions, a target-network synchronization period of 10^3 steps, gradient-norm clipping at 10, and one gradient update every two environment steps. Exploration uses ϵ -greedy action selection with ϵ decayed linearly from 1.0 to 0.05 over the first 4×10^4 steps and held at 0.05 thereafter.

5.3 Baseline (Non-Equivariant) Architecture

The baseline Q_θ is an ordinary convolutional network with no built-in symmetry. Two 3×3 convolutions ($3 \rightarrow 16 \rightarrow 16$ channels, stride 1, zero padding 1, ReLU activations) preserve the 8×8 spatial resolution; the resulting feature map is flattened and passed through a fully connected ReLU layer of width 128 and a linear layer to the four action values. The convolutions are translation-equivariant by construction but carry no rotational or reflectional symmetry, so the network is free, but in no way obligated, to represent the D_4 structure of the task.

5.4 Equivariant Architecture

For comparison we implement a D_4 -equivariant Q -network as a group-equivariant CNN in the regular representation, following the group-convolution construction of [3] (the discrete special case of the steerable framework of [4]). A lifting layer maps the input image to eight feature maps indexed by the elements of D_4 by convolving with the eight transformed copies of a shared filter bank; a group-convolution layer then maps regular-representation features to regular-representation features, transforming each filter spatially by g and permuting its group axis by the corresponding Cayley shift. After two such layers and a global spatial average, the eight per-group-element feature vectors transform by the regular representation of D_4 . A shared channel-to-scalar map produces one scalar per group element, and a final *coset pooling* step averages these scalars over the cosets of the cardinal-action orbit, yielding four action values that satisfy $Q(g \cdot s, g \cdot a) = Q(s, a)$ exactly. With an orbit-tied output bias, the resulting policy is D_4 -equivariant by construction in the sense of Equation (4).

We confirm this numerically: on random states the equivariance error (Definition 2, defined in Section 6) of the equivariant network is $\approx 9 \times 10^{-9}$ (machine zero), while an untrained baseline network of the same size already shows a nonzero error. The two architectures are matched in depth and trained with the identical DQN configuration of Section 5.2, so differences in measured equivariance are attributable to the architectural constraint rather than to the optimizer or schedule.

5.5 Training and Checkpointing

Agents are trained for up to 1.2×10^5 environment steps. To support the through-training analysis of Section 6, we save model checkpoints on a fixed schedule (at 0, i.e. random initialization, and at $10^3, 2.5 \times 10^3, 5 \times 10^3, 10^4, 2 \times 10^4, 4 \times 10^4, 6 \times 10^4$ steps and beyond), so that the equivariance metric can be evaluated on the *same* agent at successive stages of learning. Figure 1 shows the baseline learning curve: food consumption per episode rises steadily as ϵ decays and plateaus once exploration reaches its floor. Evaluated greedily over 200 episodes, the trained baseline eats 9.4 food items per episode on average (maximum 23), confirming that it learns a competent, non-trivial Snake policy. This is the regime in which asking whether it is *also* equivariant is meaningful.

5.6 Evaluation Protocol

We report two kinds of measurement. *Task performance* is the mean number of food items eaten per episode under the greedy policy, averaged over 200 independent episodes. *Equivariance* is the error $\mathcal{E}_D(\pi; D_4, \mathcal{T})$ of Definition 2, computed for both the total-variation and symmetric-KL divergences. Following Remark 3, the test set $\mathcal{T}$ is a fixed reference set of $|\mathcal{T}| = 1000$ uniformly sampled valid states that is *closed under the group action*: we draw base states and add all eight of their D_4 images, so the sampling distribution is itself D_4 -invariant and the metric cannot conflate policy asymmetry with sampling asymmetry. The softmax that converts Q -values to a distribution uses a small additive smoothing constant for the KL divergence to avoid division by zero. Because $\mathcal{T}$ is fixed across checkpoints, equivariance errors are directly comparable across training stages and between the two architectures.

6 Measuring Equivariance in Learned Policies

The theory of Section 3, instantiated for Snake in Corollary 5, establishes that an equivariant optimal policy *exists*. It says nothing about whether an RL algorithm training a non-equivariant

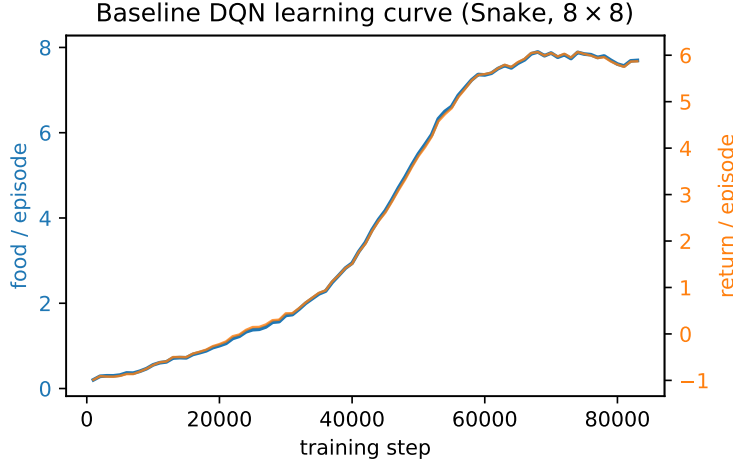


Figure 1: Baseline DQN learning curve on 8×8 Snake. Food eaten per episode (left axis) and episodic return (right axis) versus training step, averaged over a trailing window of episodes. Training converges to a competent policy before the equivariance analysis of Section 6.

network will *recover* an equivariant policy. This section addresses that empirical question.

We define a metric quantifying how equivariant a given policy is, then apply it to policies trained by standard DQN throughout training. The metric is stated for a general finite group G ; in the experiments $G = D_4$.

6.1 Equivariance Error

Given a policy $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$, we define its equivariance error with respect to a finite group G as follows. For each $g \in G$ and each state s , the policy is equivariant at (s, g) when $g \cdot \pi(s) = \pi(g \cdot s)$, i.e. when the distributions $\pi(g \cdot s)$ and $g \cdot \pi(s)$ coincide. We measure deviation from equivariance using a divergence between these distributions.

We use two divergences:

- **Total variation distance:** $\text{TV}(\mu, \nu) = \frac{1}{2} \sum_{a \in \mathcal{A}} |\mu(a) - \nu(a)|$, which lies in $[0, 1]$.
- **Symmetric KL divergence:** $\text{KL}_{\text{sym}}(\mu, \nu) = \frac{1}{2} (\text{KL}(\mu \| \nu) + \text{KL}(\nu \| \mu))$, where $\text{KL}(\mu \| \nu) = \sum_a \mu(a) \log(\mu(a)/\nu(a))$. (We use small smoothing to avoid divisions by zero in practice.)

Definition 2 (Equivariance error). Given a policy π , a divergence D on $\Delta(\mathcal{A})$, a finite group G , and a finite set of test states $\mathcal{T} \subset \mathcal{S}$, the *equivariance error* of π is

$$\mathcal{E}_D(\pi; G, \mathcal{T}) := \frac{1}{|\mathcal{T}| \cdot (|G| - 1)} \sum_{s \in \mathcal{T}} \sum_{g \in G \setminus \{e\}} D(\pi(g \cdot s), g \cdot \pi(s)). \quad (21)$$

We exclude the identity element e from the average because the contribution at $g = e$ is identically zero. The metric is zero iff π is equivariant on the test set $\mathcal{T}$.

Remark 2 (Choice of divergence). The TV distance is bounded and easy to interpret as a fraction of probability mass that disagrees. The KL divergence is unbounded but more sensitive to differences on low-probability actions. We report both.

Remark 3 (Choice of test states). The choice of test set $\mathcal{T}$ matters. Two natural choices: (i) states sampled from the policy’s own visitation distribution, which measures equivariance where the policy “actually goes,” or (ii) states sampled from a fixed reference distribution covering the state space more broadly. We use the latter (uniformly sampled valid states with various snake lengths and food positions) to allow comparison across policies at different training stages. To avoid the metric conflating policy asymmetry with sampling asymmetry, we draw $\mathcal{T}$ from a G -invariant distribution; equivalently, we close $\mathcal{T}$ under the group action so that $s \in \mathcal{T} \Rightarrow g \cdot s \in \mathcal{T}$ for all $g \in G$.

6.2 Baseline Behavior on a Single Checkpoint

We first apply the equivariance metric to the fully-trained baseline DQN agent of Section 5 (the policy is the softmax over its Q -values). On the group-closed reference set of $|\mathcal{T}| = 1000$ states, the trained baseline has

$$\mathcal{E}_{\text{TV}} = 0.160, \quad \mathcal{E}_{\text{KL}_{\text{sym}}} = 0.181. \quad (22)$$

To calibrate these numbers we evaluate two reference policies on the same test set. A uniformly random policy has $\mathcal{E}_{\text{TV}} = 0$: it is constant across states and actions and so is trivially equivariant. A degenerate policy that always chooses “up” has $\mathcal{E}_{\text{TV}} = 0.857$, close to the maximal value of 1; this is the relevant notion of “maximally asymmetric,” since the equivariant counterpart of “always up” would rotate its preferred action with the state, whereas “always up” does not. The trained baseline sits between these extremes, with a substantial 16% average disagreement in probability mass between $\pi(g \cdot s)$ and $g \cdot \pi(s)$: far from the random policy’s perfect equivariance, yet far from maximal asymmetry. The through-training and per-element breakdowns below place this value in context.

6.3 Equivariance Through Training

We measure $\mathcal{E}_{\text{TV}}$ and $\mathcal{E}_{\text{KL}_{\text{sym}}}$ at checkpoints saved every K training steps. Two questions of interest:

1. Does equivariance increase over training? That is, does the policy become *more* equivariant as it improves at the task?
2. If so, does it plateau, and at what value? Does the baseline ever achieve perfect equivariance?

Figure 2 plots $\mathcal{E}_{\text{TV}}$ and $\mathcal{E}_{\text{KL}_{\text{sym}}}$ against the training step, evaluated on the fixed test set at each saved checkpoint.

The observed pattern is the *opposite* of the naive expectation that an untrained network is maximally asymmetric and grows more equivariant with training. At initialization the equivariance error is small ($\mathcal{E}_{\text{TV}} \approx 0.03$), but for a revealing reason: the untrained network produces near-constant Q -values, so its softmax policy is close to uniform, and a near-uniform policy is trivially close to equivariant regardless of architecture. As ϵ decays and the policy becomes confident and sharply state-dependent, the absence of any architectural symmetry constraint begins to bite: the error jumps within the first few thousand steps and then fluctuates in a band around $\mathcal{E}_{\text{TV}} \in [0.11, 0.16]$, settling near 0.16 at convergence. Answering the two questions posed above: (1) equivariance does *not* increase with training, and if anything the trained policy is *less* equivariant than the random initialization once the confounding effect of near-uniform initial outputs is accounted for; and (2) the curve plateaus well above zero. The non-equivariant baseline never recovers the equivariant optimum guaranteed by Corollary 5; it converges to a competent but persistently asymmetric policy.

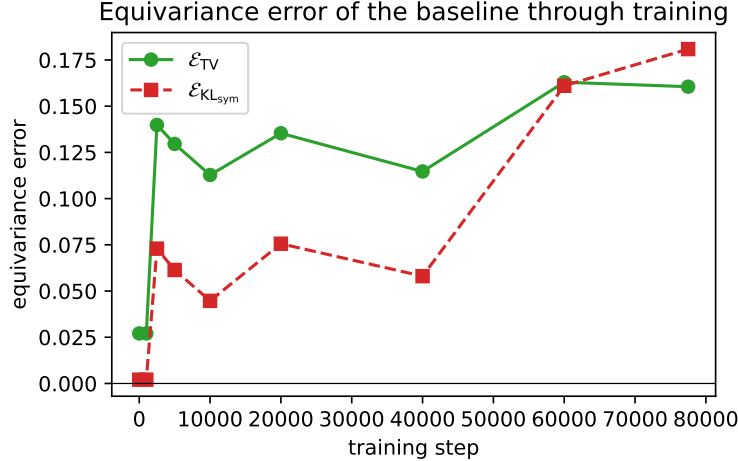


Figure 2: Equivariance error of the baseline DQN through training, on the fixed group-closed test set. The error is near zero at initialization, where an untrained network outputs near-constant Q -values, so the softmax policy is near-uniform and *trivially* near-equivariant. It then rises sharply as the policy sharpens into confident, state-dependent decisions, and plateaus around $\mathcal{E}_{TV} \approx 0.12$ – 0.16 rather than decaying to zero.

6.4 Breakdown by Group Element

The equivariance error in Definition 2 averages over the seven non-identity elements of D_4 . We can also report per-element errors,

$$\mathcal{E}_D(\pi; g, \mathcal{T}) := \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} D(\pi(g \cdot s), g \cdot \pi(s)), \quad (23)$$

to identify which symmetries are easier or harder for the network to learn implicitly.

A natural hypothesis is that 180 rotation (r^2) is easier than 90 rotation (r), because r^2 commutes with the implicit translation symmetry of CNN convolutions in a more constrained way. Figure 3 shows the per-element errors for the trained baseline. The hypothesis is *not* borne out: the error is strikingly uniform across all seven non-identity elements, ranging only over $\mathcal{E}_{TV}(\pi; g) \in [0.152, 0.167]$, and r^2 (0.165) is in fact marginally *higher* than r (0.160), not lower. The four reflections and the three nontrivial rotations are violated to essentially the same degree. This suggests that the baseline’s asymmetry is not the failure of one particular symmetry that the convolutional structure handles specially, but a broad, roughly isotropic departure from D_4 -equivariance, consistent with the policy simply having learned eight only-loosely-related responses to the eight orientations of a configuration.

6.5 Failure Modes

Beyond aggregate statistics, we examine the specific states where the policy is most asymmetric. For each state we measure its worst-case violation $\max_{g \neq e} TV(\pi(g \cdot s), g \cdot \pi(s))$; the mean over the test set is 0.27, the 95th percentile is 0.68, and the maximum approaches 1. Figure 4 visualizes the three most asymmetric states. In each, the policy commits confidently to one action for s (e.g. “right”) but, on the transformed state $g \cdot s$, commits just as confidently to an action that is *not* the corresponding transform of that choice, so the pulled-back distribution $g^{-1} \cdot \pi(g \cdot s)$ disagrees

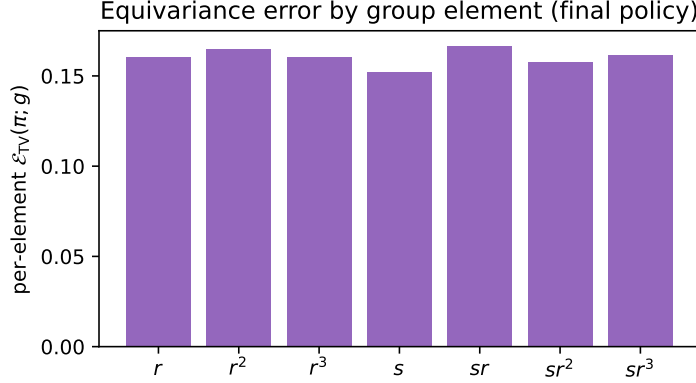


Figure 3: Per-element equivariance error $\mathcal{E}_{TV}(\pi; g)$ of the trained baseline for the seven non-identity elements of D_4 (three rotations, four reflections). The error is nearly uniform across elements; r^2 is not easier than r .

almost completely with $\pi(s)$ ($TV \approx 0.95$). These are not cases of mild probabilistic drift but of the network returning a qualitatively different decision for two configurations that are exact symmetric copies of one another.

We also test the three hypotheses about which states are hardest, comparing mean worst-case asymmetry across state subsets. (i) *Boundary*. States with the head on the grid boundary are *less* asymmetric than interior states (0.24 vs. 0.30), the opposite of the conjecture: near a wall, several moves are clearly fatal, which constrains the policy and incidentally makes it more consistent across orientations. (ii) *Snake length*. This is the dominant factor: worst-case asymmetry correlates positively with snake length ($r = 0.53$), and long snakes (above the median length) are markedly more asymmetric than short ones (0.36 vs. 0.20). Longer bodies require integrating information across more of the grid, and the unconstrained network’s responses to that longer-range structure diverge more across orientations. (iii) *Food geometry*. States with the food on a diagonal relative to the head are *less* asymmetric than states with axially-aligned food (0.26 vs. 0.34), again contrary to the conjecture. Of the three, only the snake-length effect is confirmed; the boundary and diagonal effects run opposite to the stated hypotheses.

6.6 Summary

The baseline learns competent play but not equivariance: its error rises from initialization to a plateau of $\mathcal{E}_{TV} \approx 0.16$, spread evenly across D_4 and worst on long snakes, so it never recovers the equivariant optimum Corollary 5 guarantees, motivating the equivariant architecture of Section 7.

7 Comparison with Equivariant Architectures

Section 6 showed that a standard DQN agent learns a competent but persistently non-equivariant policy. Corollary 3 guarantees that an equivariant optimum exists, and equivariant architectures realize the restriction to equivariant policies *by construction*. We now train the D_4 -equivariant network of Section 5.4 with the identical DQN configuration and compare it against the baseline along two axes: how equivariant the learned policy is, and how well and how quickly it plays.

training budget	greedy food / episode		equivariance $\mathcal{E}_{\text{TV}}$ (equ.)
	baseline CNN	D_4 -equivariant CNN	
2×10^4 steps	2.4	9.5	$\approx 10^{-8}$
4×10^4 steps	7.3	17.1	$\approx 10^{-8}$
baseline best (6×10^4)	9.9	–	–
parameters	134,484	6,262	

Table 1: Greedy task performance at matched training budgets (mean food per episode over 120 episodes), and parameter counts. The equivariant network reaches the baseline’s fully-trained performance in a third of the steps, with roughly $21\times$ fewer parameters, while remaining exactly equivariant.

7.1 Equivariance by Construction

By design the equivariant agent satisfies $Q(g \cdot s, g \cdot a) = Q(s, a)$ exactly, so its softmax policy is exactly equivariant regardless of its weights. Figure 5 confirms this empirically and contrasts it with the baseline: across every saved checkpoint the equivariant agent’s error is $\mathcal{E}_{\text{TV}} \approx 10^{-8}$ (machine zero, attributable entirely to floating-point round-off), while the baseline fluctuates between 0.03 and 0.16. The gap is roughly seven orders of magnitude and, crucially, is *constant over training*: the equivariant agent is exactly equivariant at random initialization, at convergence, and everywhere in between. There is no learning required to acquire the symmetry and no residual asymmetry left at the end. This is the architectural payoff of Corollary 5: rather than hoping the optimizer discovers the equivariant optimum, the network is constrained to the equivariant policy class from the outset.

7.2 Task Performance and Sample Efficiency

Equivariance would be a hollow victory if it cost task performance. It does not. Figure 6 plots food consumed per episode during training (under the shared ϵ -greedy schedule, so the two curves are directly comparable), and Table 1 reports greedy evaluation at matched training budgets. The equivariant agent is more sample-efficient at every stage and reaches a substantially higher final level of play. After 2×10^4 steps it already eats 9.5 food items per episode under greedy evaluation, matching the *fully-trained* baseline’s ≈ 9.4 –9.9 (Section 6 and Table 1), and after 4×10^4 steps it reaches 17.1, nearly double the baseline’s best. At the matched budget of 4×10^4 steps the equivariant agent outperforms the baseline by more than a factor of two (17.1 vs. 7.3).

Notably, the equivariant agent reaches this performance with roughly $21\times$ fewer parameters (6,262 vs. 134,484): weight sharing across the group is a correct prior for an exactly symmetric task, so the network never spends capacity or samples on separate responses to the eight orientations of a configuration, the failure mode diagnosed in Section 6. The offsetting cost is per-step compute, since each group-convolution evaluates eight oriented copies of every filter; we take up when this trade is worthwhile, and the limitations of the comparison, in Section 8.

8 Discussion

Existence versus learnability. The central tension of this paper is between two facts that are both true and easily conflated. Corollary 3 establishes that every finite G -symmetric MDP *admits*

an equivariant optimal policy; in Snake, an optimal player can always be made to respond to a rotated board with the rotated move. Yet Section 6 shows that a standard learning algorithm does not *recover* this structure: the baseline DQN converges to a competent but stubbornly non-equivariant policy, with an equivariance error that, far from decaying toward the guaranteed-attainable zero, grows from its near-uniform initialization and plateaus around $\mathcal{E}_{\text{TV}} \approx 0.16$. The existence theorem constrains what an optimal policy *could* look like, not what gradient descent on a symmetry-blind function class *will* find. Nothing in the DQN objective rewards equivariance, so the network spends capacity and samples learning eight loosely-coupled responses to the eight orientations of a configuration. This is visible in Section 6 as a roughly isotropic error across the group and as outright contradictory decisions on the most asymmetric states.

When equivariant architectures are worth their cost. Section 7 gives the constructive answer: an architecturally equivariant agent closes the gap by fiat, remaining exactly equivariant at every checkpoint while learning faster and playing better. Its cost is higher per-step compute, and more fundamentally a hard commitment to the symmetry being real, which is where Remark 1 becomes the operative caveat. Snake is *exactly* G -symmetric (Proposition 4), so the equivariant prior is simply correct and pays for itself many times over. In environments whose symmetry is only approximate, broken by an asymmetric goal, a boundary effect, gravity, or a direction-dependent reward, the same hard constraint forces the policy to ignore a genuine signal, and the favorable trade of Section 7 can invert. The decision to use an equivariant architecture is therefore not primarily about model size or accuracy but about how exactly the environment’s symmetry holds; our results bracket the exact-symmetry end of that spectrum.

Limitations. Several caveats temper the empirical claims. The comparison uses a single training seed per architecture; the matched-budget gaps in Table 1 are large relative to typical DQN seed variance, but we do not quantify that variance here. We study one environment (8×8 Snake) under one group (D_4), one algorithm (value-based DQN), and one action encoding (absolute cardinal directions); whether the qualitative findings (residual asymmetry that grows rather than shrinks, isotropy across the group, the length dependence) transfer to policy-gradient methods, larger grids, or other symmetry groups is open. Our equivariance metric is computed on a fixed group-closed reference distribution rather than the policy’s own visitation distribution (Remark 3), and its absolute magnitude depends on the softmax temperature used to turn Q -values into a policy; we therefore read it comparatively (across checkpoints and architectures) rather than as an absolute physical quantity. Finally, architectural equivariance is not the only route to symmetry: symmetry-augmented training (averaging the loss or the data over the group) is a softer alternative we did not evaluate, and a three-way comparison with it would sharpen the practical guidance.

Future directions. The most immediate extensions follow the limitations: multiple seeds with variance estimates, a data-augmentation baseline as a third point between the unconstrained and hard-equivariant extremes, and the second-environment study sketched in ?? to test whether the residual-asymmetry phenomenon is specific to Snake/ D_4 or general to symmetry-blind RL. On the theory side, Remark 1 points to two relaxations, compact groups via Haar-measure averaging and continuous state spaces via measure-theoretic arguments, that would extend Corollary 3 beyond the finite setting and connect it to the MDP-homomorphism framework of [1, 2]. Most interesting is the approximate-symmetry regime: quantifying how the equivariance error of the optimal policy, and the cost of imposing exact equivariance, scale with the size of the symmetry violation would turn the qualitative trade-off above into a usable design rule.

9 Conclusion

We studied symmetry in reinforcement learning at three levels. Theoretically, we developed the notion of a finite G -symmetric MDP for an arbitrary finite group and proved, in self-contained form, that its optimal value is G -invariant (Lemma 1), that the optimal policies are closed under the group action (Theorem 2), and consequently that an equivariant optimal policy always exists (Corollary 3). Concretely, we formalized Snake as an MDP, exhibited an explicit D_4 action on its states and actions, and verified in proof and in code that the resulting MDP is exactly D_4 -symmetric (Proposition 4), so the existence of an equivariant optimum is inherited for free. Empirically, we introduced an equivariance-error metric and used it to show that a standard DQN agent does *not* recover the equivariant optimum its environment guarantees: its asymmetry grows from a near-uniform initialization, plateaus well above zero, spreads roughly evenly across the group, and concentrates in longer-snake configurations where the policy makes contradictory choices on symmetric states. Finally, we showed that a D_4 -equivariant network closes this gap by construction, remaining exactly equivariant throughout training while learning faster and playing better than the baseline with an order of magnitude fewer parameters.

The takeaway is that an existence guarantee about symmetric optima and the behavior of a symmetry-blind learner are different things, and the distance between them is exactly what equivariant architectures are for. When a task’s symmetry is exact, as in Snake, building that symmetry into the network is not merely a regularizer but the correct hypothesis class, and the empirical benefits are correspondingly large. Charting how this advantage erodes as symmetry becomes approximate is, to us, the most promising direction this work opens.

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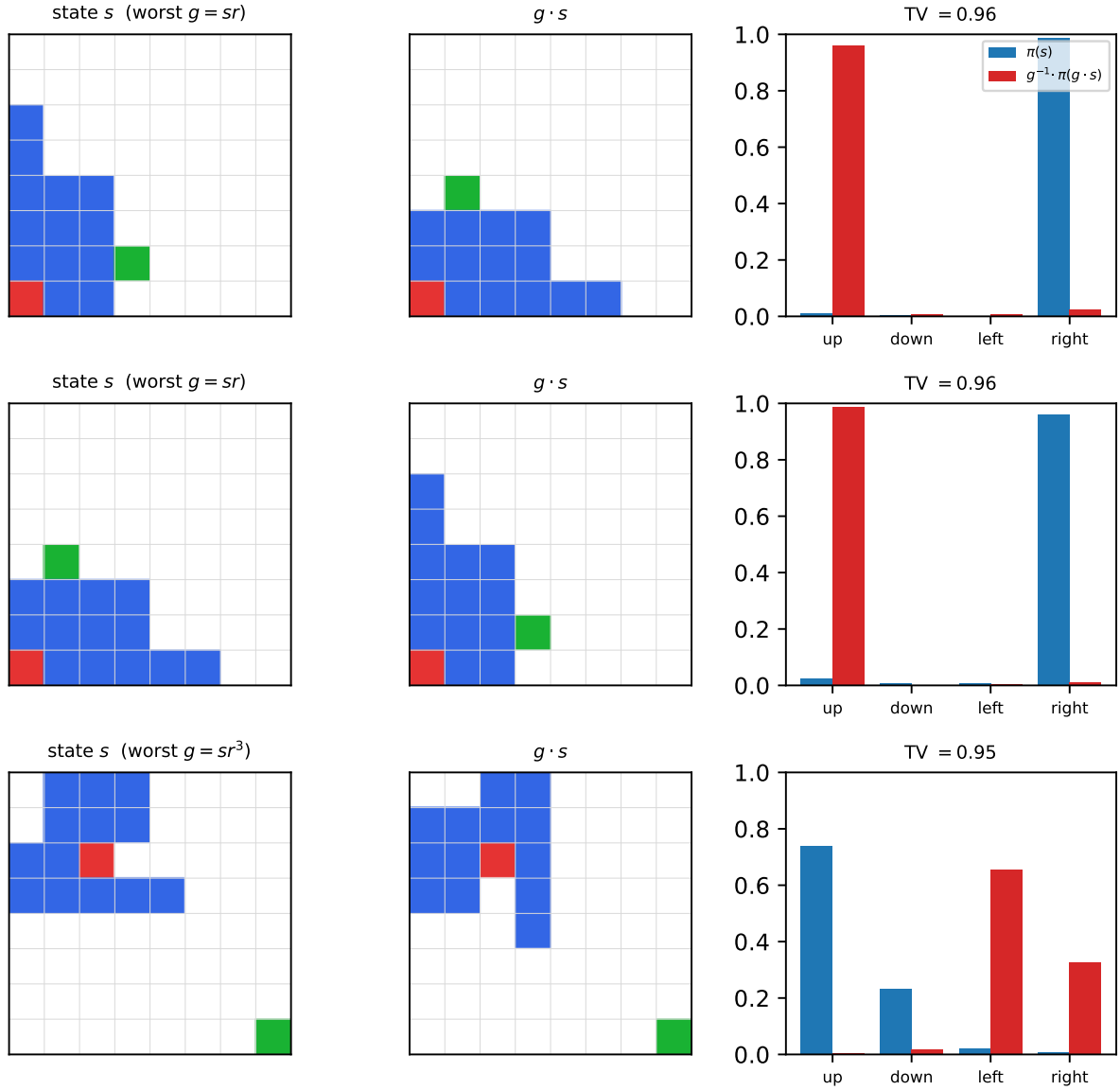


Figure 4: The three most asymmetric test states for the trained baseline. Each row shows the state s , its transform $g \cdot s$ under the worst-case element g (red: head, blue: body, green: food), and the action distributions $\pi(s)$ (blue) versus $g^{-1} \cdot \pi(g \cdot s)$ (red). Equivariance would make the two bar charts coincide; instead the policy makes near-opposite confident choices on symmetric configurations.

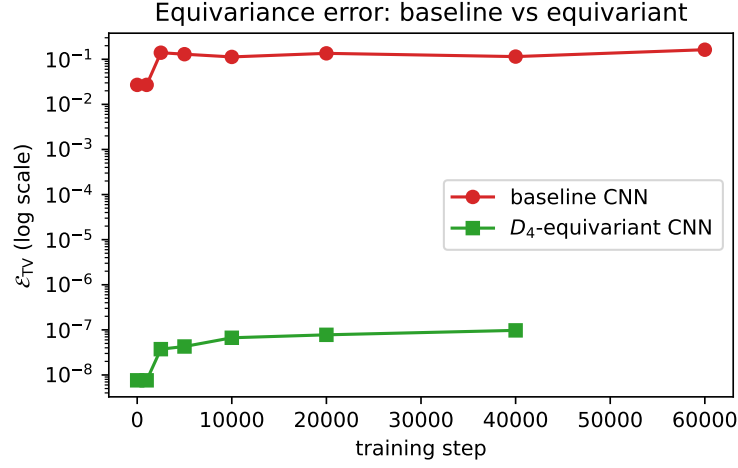


Figure 5: Equivariance error $\mathcal{E}_{TV}$ through training (log scale), baseline versus D_4 -equivariant agent, on the fixed group-closed test set. The equivariant agent is exactly equivariant ($\approx 10^{-8}$, machine zero) at every checkpoint; the baseline never falls below ≈ 0.1 .

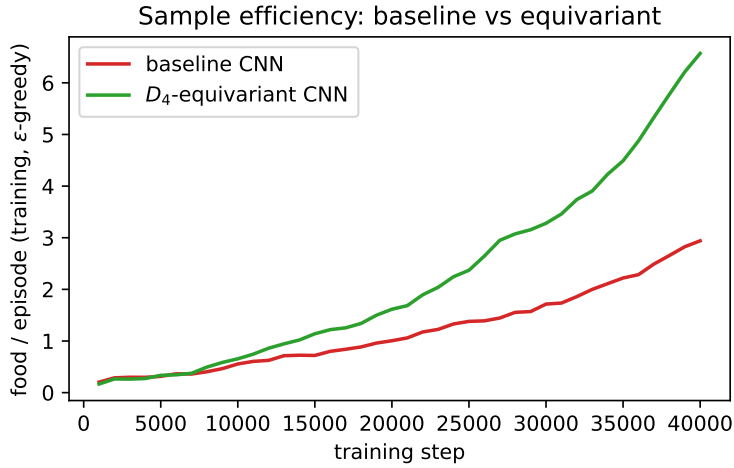


Figure 6: Food per episode through training (training-time ϵ -greedy behavior, identical exploration schedule for both agents). The D_4 -equivariant agent learns faster throughout.